Alex Morgan

alex@example.com · GitHub.com/alex-example · LinkedIn.com/in/alex-example

Education

New York University — B.S. in Computer Science Sep 2020 – May 2024

New York, NY

- Minor in Mathematics | GPA: 3.88 / 4.0 | Dean's Honor List
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Systems Organization, Database Systems, Computer Networks, Distributed Systems, Machine Learning

Research & Professional Experience

Software Engineering Intern Jun 2024 – Aug 2024

Datadog · New York, NY

- Engineered high-throughput event streaming pipelines using Go and Apache Kafka, processing 50M+ telemetry metrics daily with sub-second latency.
- Designed and deployed containerized RESTful microservices with Docker and Kubernetes, reducing API response times by 35%.
- Implemented automated end-to-end integration test suites in CI/CD pipelines, increasing branch test coverage from 78% to 93%.

Undergraduate Research Assistant Sep 2023 – May 2024

NYU Courant Machine Learning and Systems Lab · New York, NY

- Researched memory-efficient attention mechanisms and distributed training optimizations for transformer models in PyTorch.
- Profiled GPU kernel execution with NVIDIA Nsight and CUDA to identify memory bandwidth bottlenecks during multi-GPU inference.
- Developed scalable data preprocessing pipelines in Python and Ray to clean and tokenize 2TB+ multimodal benchmark datasets.

Software Engineering Intern Jun 2023 – Aug 2023

MongoDB · New York, NY

- Built full-stack telemetry dashboards using TypeScript, React, and Node.js for real-time database cluster health monitoring.
- Optimized aggregation pipeline queries for distributed document collections, improving query execution latency by 40%.
- Collaborated across an agile team of 6 engineers using Git, code reviews, and bi-weekly sprint planning.

Projects

RaftKV — Distributed Fault-Tolerant Key-Value Store Jan 2024 – Apr 2024

- Engineered a distributed key-value store in Go implementing the Raft consensus protocol for high availability and strong consistency.
- Supported leader election, log replication, dynamic cluster reconfiguration, and snapshot compaction.
- Benchmarked performance under simulated network partitions, sustaining 15,000+ requests/sec across 5-node clusters.

CloudNet — Containerized Microservices Platform Sep 2023 – Dec 2023

- Architected a scalable microservices web application using Python (FastAPI), React, PostgreSQL, and Redis caching.
- Automated continuous integration and deployment with GitHub Actions and Helm charts to a Kubernetes cluster.
- Integrated Prometheus metrics and Grafana dashboards for real-time monitoring of service latency and resource usage.

Skills

Languages:	Python, Go, C/C++, Java, TypeScript, JavaScript, SQL, Rust, Bash
Frameworks & Libraries:	React, Node.js, FastAPI, Next.js, PyTorch, Express, GraphQL, gRPC
Systems & Cloud:	Docker, Kubernetes, Linux / Unix, AWS (EC2, S3, Lambda), PostgreSQL, Redis, Apache Kafka, Git, CI/CD (GitHub Actions)
Core Concepts:	Distributed Systems, Algorithms & Data Structures, Object-Oriented Design, RESTful APIs, Microservices, Machine Learning